On the Information Transfer Rate of SPAD Receivers for Optical Wireless Communications

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***Abstract*—In this paper, the information rate of single photon avalanche diode (SPAD) receivers with active quenching circuits for optical communication systems is studied. The SPAD receiver is modeled as a discrete memoryless channel (DMC), and the information transfer rate is studied using an information theoretic approach. To assess the input-output information transfer rate of the SPAD optical receiver, a channel capacity metric is proposed which is a function of several parameters including average signal photon count, SPAD dead time, and SPAD average background counts. The proposed metric can be used for theoretical optimization of the device structure and operating conditions. Performance results are presented based on the proposed metric.**

***Index Terms*—Single Photon Avalanche Diode (SPAD), active quenching, photon counting, dead time, channel capacity.**

I. Introduction

Single Photon Avalanche Diodes (SPADs) are gaining interest in the field of optical wireless communications (OWC) owing to their high power efficiency, high sensitivity, high detection efficiency, and high timing resolution [1]–[3]. SPADs are able to closely approach quantum-limited sensitivity in the detection of weak optical signals and this has recently highlighted the potential of employing them in applications where long distance transmission requires highly sensitive receivers capable of detecting a single photon [4], [5].

SPADs are semiconductor devices with p-n junctions and are operated in avalanche breakdown photon counting mode, also termed as 'Geiger mode'. In Geiger mode, the p-n junction is biased above the breakdown voltage such that individual photons trigger an avalanche breakdown. The avalanche breakdown results in a strong output current spike which is then individually counted [6]. Every time an avalanche breakdown occurs, the output current has to be quenched by lowering the bias voltage down to or below breakdown voltage to stop the avalanche event, and in order to detect a subsequent photon, the bias voltage must be raised again. This is accomplished by a suitable electronic circuit known as 'quenching circuit'. During the quenching process, SPAD is unresponsive for a circuit specific 'dead time'. Two approaches can be followed to design a quenching circuit: passive quenching (PQ) and active quenching (AQ). The configurations of PQ and AQ circuits are presented in [7]. In general, PQ SPADs are identified as paralyzable detectors where any photon arrival occurring during the dead time is not counted but is assumed to extend the dead time period; while in an AQ SPAD device, the duration of the dead time introduced by the quenching process is constant, that is, any photon arriving during the dead time is neither counted nor has any influence on the dead time duration. Thus AQ SPADs are defined as nonparalyzable detectors and generally have higher count rates than PQ SPADs [7].

Quenching circuits directly affect the performance of the device because of the dead time they introduce for the recovering process. The relatively long dead time due to the slow recovery process, limits counting rates, and results in a considerable count loss relative to an ideal dead time-free photon counting detector, however second order effects such as afterpulsing, prevent arbitrarily short dead time. The dead time of commercially available SPAD receivers vary in the range of a few nanoseconds to tens of nanoseconds, causing significant losses for communication links with slot widths of the same order. This has motivated recent studies on characterizing losses due to the SPAD dead time [8], [9]. Furthermore, while dead time confines the highest measurable count rate, background noise limits the low count rates.

To date, there has been extensive work on designing SPAD receivers for a number of applications including three-dimensional imaging [10], quantum key distribution [11], and deep space laser communications [12], and much effort has been devoted to optimizing the SPAD key performance metrics (e.g. detection efficiency, dead time, dark count rate, timing jitter, etc.) to improve its performance for communication purposes [1], [2], [4], [13]. In all the aforementioned articles, the impact of SPAD dead time and dark current on the performance of the systems have been experimentally explored and it is reported that due to these non-idealities the output rate is limited. Currently, efforts are underway to integrate SPAD elements directly with quenching circuitry using standard digital CMOS technology. Aiming to achieve higher output rates for optical communication applications, SPAD arrays fully integrated with the CMOS technology have been implemented. Nevertheless, highest achievable data rates are still restricted to a few tens of MHz. The effect of dead time can be mitigated by improving the performance of active quenching circuits and by device multiplexing techniques. For example in [2], a reconfigurable 32×32 SPAD array receiver integrated in standard 130 nm CMOS is presented for optical links and the properties of the receiver for optical communications are investigated, and a data rate of 100 MHz is reported.

While SPAD detectors remain an active area of develop-

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ment, there does not exist enough theoretical analysis which would specify the performance limitations of these receivers. In [8], we studied the effect of dead time on the photocount statistics of a single SPAD receiver where the probability of detection and the error probability in an optical communication system were analyzed. Also, in [9], the counting distribution and count rate of SPAD arrays were discussed. The aim of this study is to investigate how the maximum achievable information transfer rate of a SPAD receiver is limited by non-ideal behavior of an AQ circuit.

The rest of this paper is organized as follows. In Section II, a brief review of SPAD photocounting statistics in the presence of dead time is given, and the impact of dead time on the maximum achievable count rate of the SPAD receiver is discussed. In Section III, the modeling of a SPAD device as a discrete memoryless channel (DMC) with binary inputs is presented, and the channel capacity metric is introduced to assess the information transfer rate of the SPAD receiver. In Section IV, the numerical results are presented, and discussions on the effect of various parameters on the information transfer rate of the SPAD receiver are provided. Finally, concluding remarks are given in Section V.

II. SPAD Counting Statistics

In the following, the key statistical characteristics of AQ SPADs are summarized:

1) Probability Mass Function (PMF): In the absence of SPAD dead time, the detection of photon arrival events can be modeled as a Poisson arrival process for which the probability of detecting k photons over a time period of $[0, T_\mathrm{b})$ is given by [14], [15]:

$$p_0(k;\lambda) = \frac{(\lambda T_\mathrm{b})^k e^{-\lambda T_\mathrm{b}}}{k!} \,, \tag{1}$$

where the constant λ is the average photon arrival rate (in photons/sec), and hence, λT_b is the average number of photons arriving at the SPAD during the observation time of T_b seconds. The photocount rate λ is related to the optical signal power by [14], [15]:

$$\lambda = \frac{\eta_\mathrm{QE} P_\mathrm{s}}{h\nu} \,, \tag{2}$$

where η_QE is the quantum efficiency of SPAD; P_s denotes the incident optical signal power; h is the Planck's constant; and ν represents the optical signal frequency.

When the SPAD dead time is considered, however, the photon counts are no longer Poisson distributed. Assuming that the SPAD detector is ready to operate at the beginning of the counting interval, the maximum observable counts during this period is $k_\mathrm{max} = \lfloor T_\mathrm{b}/\tau \rfloor + 1$, where τ is the dead time and $\lfloor x \rfloor$ denotes the largest integer that is smaller than x. In [8], it was shown that the probability of k photons being detected during the time interval of $[0, T_\mathrm{b})$ is given by:

$$p_K(k;\lambda) = \begin{cases} \sum_{i=0}^{k} \psi(i,\lambda_k) - \sum_{i=0}^{k-1} \psi(i,\lambda_{k-1}) & k < k_\mathrm{max} \\ 1 - \sum_{i=0}^{k-1} \psi(i,\lambda_{k-1}) & k = k_\mathrm{max} \\ 0 & k > k_\mathrm{max} \end{cases} \tag{3}$$

where $\lambda_k = \lambda(T_\mathrm{b} - k\tau)$; and the function $\psi(.)$ is defined as:

$$\psi(i,\lambda) = \frac{\lambda^i e^{-\lambda}}{i!} \,. \tag{4}$$

2) First and second moments: The mean and variance of the photocount distribution in (3) are derived as:

$$\mu_K = k_\mathrm{max} - \sum_{k=0}^{k_\mathrm{max}-1} \sum_{i=0}^{k} \psi(i,\lambda_k), \tag{5}$$

$$\sigma_K^2 = \sum_{k=0}^{k_\mathrm{max}-1} \sum_{i=0}^{k} (2k_\mathrm{max} - 2k - 1)\psi(i,\lambda_k) - \left(\sum_{k=0}^{k_\mathrm{max}-1} \sum_{i=0}^{k} \psi(i,\lambda_k) \right)^2 . \tag{6}$$

It can be verified that as dead time goes to zero, the PMF in (3) approaches the ideal Poisson distribution. In such a case, the limiting relations $\lim_{\tau\to 0} \mu_K = \lambda T_\mathrm{b}$ and $\lim_{\tau\to 0} \sigma_K^2 = \lambda T_\mathrm{b}$ can also be confirmed, where λT_b is the mean of the ideal Poisson distribution.

3) Effective count rate: For a SPAD detector, background counts and dead time losses limit the minimum and maximum achievable count rates. While dead time puts restrictions on the highest measurable count rate, background noise limits the lowest count rate. The maximum count rate of commercial SPADs is restricted to a few MHz, and the main reason of this low bandwidth response is the slow quenching process after a detection event. There is another source of counting errors called afterpulsing, and it refers to avalanche events that originate from the emission of carriers trapped in the multiplication region during previous avalanche events. The afterpulsing phenomenon, prevent arbitrarily short dead time, and hence, limits the maximum achievable count rate indirectly.

According to the nonparalyzable dead time model for single-photon detectors, the relationship between the actual count rate (i.e., incoming photon rate), λ, and the effective count rate (i.e., observed rate), λ', is given by [16], [17]:

$$\lambda' = \frac{\lambda}{1 + \lambda\tau} \,, \tag{7}$$

where τ is the nonparalyzable dead time for an AQ SPAD. Note that, the maximum predicted count rate would be equal to $1/\tau$, assuming a time interval of T_b seconds. This is termed as 'saturation count rate', which means that an AQ SPAD is not able to reach count rates higher than this value.

III. Channel Capacity

The reliability performance of SPAD receivers is specified by the error and detection probabilities. The maximum achievable data rate is another key metric to characterize SPAD receivers, and this determines whether or not it is suitable for optical communication systems. To address this question, the SPAD detector can be assumed as a communication channel, and in this case the channel capacity would be the appropriate metric to assess the input-output information transfer rate of the SPAD receiver. In this context, the relevance of the channel capacity as a performance metric is clear. As long as the transfer rate of information through the SPAD device is less than the SPAD capacity, it is possible to make the error probability arbitrarily small with proper modulation and coding. In the following, the capacity of the SPAD receiver as a communication channel is investigated.

A. Discrete Memoryless Channel (DMC)

Here, the SPAD receiver is modeled as a DMC which comprises a finite input alphabet $\mathcal{X} = \{x_1, x_2, ..., x_N\}$, and a finite output alphabet $\mathcal{Y} = \{y_1, y_2, ..., y_M\}$. Let $p_X(x_i)$ be the probability that the input X takes the value x_i, for $i = 1$, 2, ..., N, and let $p_Y(y_j)$ be the probability that the output Y takes the value y_j, for $j = 1$, 2, ..., M. Define $p_{XY}(x_i, y_j)$ as the joint probability of the random variables X and Y evaluated for x_i and y_j, and also define $p_{Y|X}(y_j|x_i)$ as the conditional probability, which expresses the probability of observing the output symbol y_j given the input symbol x_i. The mutual information $I(x_i; y_j)$ between the events $X = x_i$ and $Y = y_j$ is given by [18], [19]:

$$I(x_i; y_j) = \log\left[p_{Y|X}(y_j|x_i)\right] - \log\left[p_Y(y_j)\right] = \log\left[\frac{p_{Y|X}(y_j|x_i)}{p_Y(y_j)}\right], \tag{8}$$

where the logarithm is taken to the base 2. The average mutual information between the input X and the output Y is then:

$$I(X;Y) = \sum_{x_i}\sum_{y_j} p_{XY}(x_i, y_j) \log\left[\frac{p_{Y|X}(y_j|x_i)}{p_Y(y_j)}\right], \tag{9}$$

which can be rewritten for simplicity as:

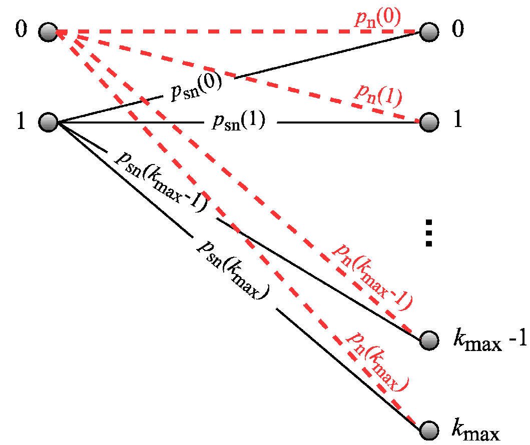
$$\begin{aligned} I(X;Y) &= \sum_{x}\sum_{y} p_{XY}(x, y) \log\left[\frac{p_{Y|X}(y|x)}{p_Y(y)}\right] \\ &= \sum_{x}\sum_{y} p_X(x) p_{Y|X}(y|x) \log\left[\frac{p_{Y|X}(y|x)}{p_Y(y)}\right], \end{aligned} \tag{10}$$

where x and y represent the values of the random variables X and Y. Also, the marginal PMF $p_Y(y)$ is given by:

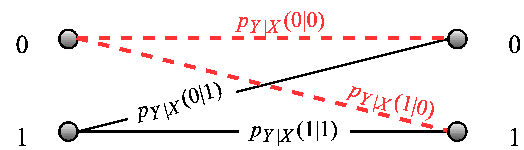
$$p_Y(y) = \sum_{x} p_X(x) p_{Y|X}(y|x) . \tag{11}$$

Therefore, if the conditional probability $p_{Y|X}(y|x)$ is given for the channel, $I(X;Y)$ in (10) can be maximized with respect to $p_X(x)$ to obtain the channel capacity as:

$$C = \max_{p_X(x)} I(X;Y) . \tag{12}$$



(a) Soft Decisions



(b) Hard Decisions

Fig. 1. DMC modeling of a SPAD receiver.

Assuming a binary signaling case for the SPAD receiver, the input X takes the value 1 when a signal is present, and 0 when there is no incoming photon. Two categories may be considered depending on the SPAD output and the type of information provided to the decoder by the SPAD receiver:

1) The SPAD receiver makes no explicit symbol decision, but passes on the slot counts directly to the decoder. This process is termed as "soft decision". The soft decision capacity is expressed as a function of the channel conditional probability $p_{Y|X}(y|x)$.
2) The SPAD is designed such that it makes estimates of each input symbol, passing on these estimates to the decoder. This process is termed as "hard decision". The "hard decision" capacity is expressed as a function of the probability of symbol errors.

Fig. 1 shows the DMC model for the SPAD receiver with soft and hard outputs. In the following, the soft and hard decision capacities for the SPAD receiver with dead time are discussed.

B. Soft Decision Capacity

Define $\lambda_{\rm s}$ as the average incoming signal photon rate. Note that the SPAD receiver has an internal source of Poisson distributed noise, due to background counts such as dark counts and afterpulsing. Let the average photon rate from the background noise be denoted by $\lambda_{\rm n}$. Therefore, when $X = 0$, the average number of photons in the time interval $[0, T_{\rm b})$ is $K_{\rm n} = \lambda_{\rm n} T_{\rm b}$, and if $X = 1$, the average number of received photons is $K_{\rm s} + K_{\rm n} = (\lambda_{\rm s} + \lambda_{\rm n}) T_{\rm b}$. According to (3), the probabilities that exactly k photons are counted by the SPAD receiver in the counting interval of $T_{\rm b}$ seconds, denoted as $p_{\rm n}(k)$ and $p_{\rm sn}(k)$, for the corresponding input values of $X=0$ and $X=1$, are given by:

$$\begin{aligned} p_{\rm n}(k) &= p_K(k;\lambda_{\rm n})\ , \\ p_{\rm sn}(k) &= p_K(k;\lambda_{\rm s}+\lambda_{\rm n})\ . \end{aligned} \tag{13}$$

For the soft decision case, the output of the SPAD receiver, Y, is equal to the number of counts registered during the counting interval of $[0,T_{\rm b})$; thus Y can take all non-negative integer values 0, 1, 2, ..., $k_{\max}$. Furthermore, it follows that:

$$\begin{aligned} p_{Y|X}(y|0) &= p_{\rm n}(k)\ , \\ p_{Y|X}(y|1) &= p_{\rm sn}(k)\ . \end{aligned} \tag{14}$$

Assuming $p_X(x=1)=q$ and $p_X(x=0)=1-q$, the average mutual information $I(X,Y)$ given in (10) is expressed as:

$$\begin{aligned} I(X;Y) = \sum_{k=0}^{k_{\max}} \Bigg\{ &(1-q)\ p_{\rm n}(k) \log\left[\frac{p_{\rm n}(k)}{p_Y(k)}\right] \\ &+ q\ p_{\rm sn}(k)\log\left[\frac{p_{\rm sn}(k)}{p_Y(k)}\right]\Bigg\}\ , \end{aligned} \tag{15}$$

where $p_Y(k)$ is given using (11) as:

$$p_Y(k) = (1-q)\ p_{\rm n}(k) + q\ p_{\rm sn}(k)\ . \tag{16}$$

With (15) and (16), $I(X;Y)$ can be calculated for any given value of q, and thus the channel capacity C can be found using (12). For the SPAD receiver considered here, the channel capacity is a function of source statistics, SPAD dead time, and average signal and background counts.

C. Hard Decision Capacity

In this case, the output Y may take only two values, 0 and 1, as determined by a likelihood-ratio test. Thus, the conditional probabilities, $p_{Y|X}(y|x)$, for the four possible combinations of the binary input-output pair, (x,y), can be calculated as:

$$p_{Y|X}(0|0) = \Pr\left\{k \le k_{\rm TH}|x=0\right\} = \sum_{k=0}^{k_{\rm TH}} p_{\rm n}(k), \tag{17}$$

$$p_{Y|X}(1|0) = 1 - p_{Y|X}(0|0), \tag{18}$$

$$p_{Y|X}(0|1) = \Pr\left\{k \le k_{\rm TH}|x=1\right\} = \sum_{k=0}^{k_{\rm TH}} p_{\rm sn}(k), \tag{19}$$

$$p_{Y|X}(1|1) = 1 - p_{Y|X}(0|1), \tag{20}$$

where $k_{\rm TH}$ denotes the optimum threshold for maximum likelihood detection, which is given by:

$$k_{\rm TH} = \frac{K_{\rm s} + \ln\left(\frac{1-q}{q}\right)}{K_{\rm s}\delta + \ln\left(1+\frac{K_{\rm s}}{K_{\rm n}}\right)}\ , \tag{21}$$

and $\delta = \tau/T_{\rm b}$ is the dead time ratio. Based on (17)–(20), and using (10), the average mutual information is calculated as:

$$\begin{aligned} I(X;Y) = \sum_{y=0}^{1} \Bigg\{ &(1-q)\ p_{Y|X}(y|0)\log\left[\frac{p_{Y|X}(y|0)}{p_Y(y)}\right] \\ &+\ q\ p_{Y|X}(y|1)\log\left[\frac{p_{Y|X}(y|1)}{p_Y(y)}\right]\Bigg\}\ . \end{aligned} \tag{22}$$

Note that, $p_{Y|X}(y|x)$ in (22) is a function of *a priori* probability q, since it depends on $k_{\rm TH}$, and $k_{\rm TH}$ given in (21) is a function of q.

IV. Numerical Results

Figs. 2–4 present performance results for the average mutual information, $I(X;Y)$, of the SPAD receiver as a function of *a priori* probability, q, for both soft decision and hard decision categories, based on (15) and (22) from Section III. The solid curves refer to the SPAD receiver with soft decision output, and the dashed curves represent the SPAD receiver with hard decision output. In Fig. 2 the effect of average signal count, $K_{\rm s}$, is studied for $K_{\rm s}$ = 5, 15, 30, with an average background count of $K_{\rm n}=10$ and a dead time ratio of δ = 0.05. In Fig. 3 the effect of average background count, $K_{\rm n}$, is studied for $K_{\rm n}$ = 5, 10, assuming $K_{\rm s}=20$ and δ = 0.05. In Fig. 4 the effect of the dead time ratio is studied for δ = 0, 0.05, 0.1, assuming $K_{\rm s}=20$ and $K_{\rm n}=10$. These results indicate that, the average mutual information decreases with decreasing signal mean (see Fig. 2), increasing background counts mean (see Fig. 3), and increasing dead time (see Fig. 4). It is observed that the peak value of each curve which represents the channel capacity, occurs at or near q = 0.5 in most cases.

An obvious distinction between the two types of SPAD receivers is that the hard decision SPAD exhibits discontinuities in the $I(X;Y)$ curves. This is because of discrete nature of the decision threshold $k_{\rm TH}$ which jumps from one integer to the next at certain values of q. Furthermore, SPAD receivers with soft outputs result in higher average mutual information compared with the receivers with hard outputs. Thus, the hard decision SPAD gives a lower capacity than the SPAD receiver with soft outputs in all cases. This is because the slot counts provide additional information to the decoder in the soft decision case. Since both input and output take the values 0 and 1, a hard output SPAD receiver may be considered as an asymmetric binary channel with varying error transition probability, i.e. $p_{Y|X}(1|0)$ and $p_{Y|X}(0|1)$. Therefore, the channel capacity would not exceed 1.

Figs. 5–7 provide performance results for soft and hard decision capacities of the SPAD receiver, based on (15) and (22) from Section III. In Fig. 5, it is shown how the capacity is affected by the average signal count, $K_{\rm s}$. In this figure, an average background count of $K_{\rm n}=10$ and two dead time ratios of δ = 0.1, 0.001 are assumed. As shown in Fig. 2, higher average signal counts result in higher capacity values for both soft decision and hard decision cases. However, the capacity cannot exceed 1 and the minimum average signal count required for achieving the maximum capacity depends

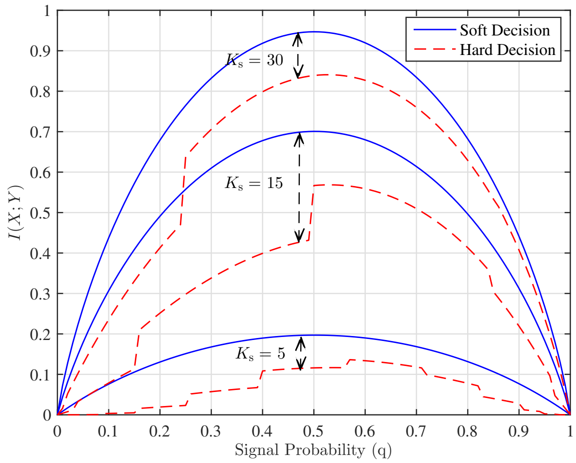


Fig. 2. Average mutual information, $I(X;Y)$, versus *a priori* signal probability, q, with $\delta = 0.05$, $K_{\mathrm{n}} = 10$.

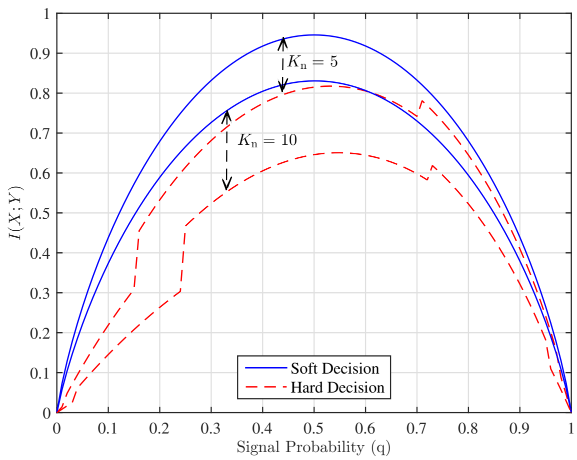


Fig. 3. Average mutual information, $I(X;Y)$, versus *a priori* signal probability, q, with $\delta = 0.05$, $K_{\mathrm{s}} = 20$.

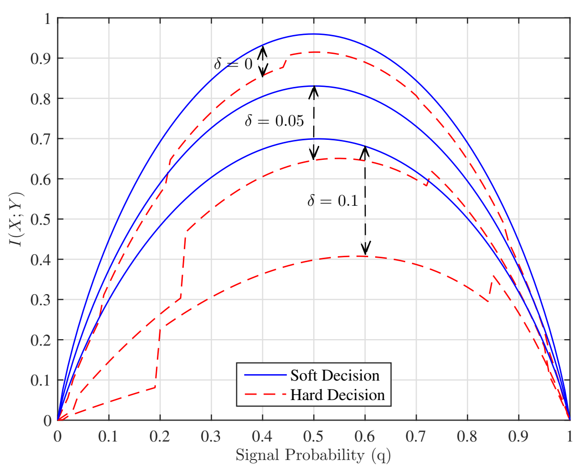


Fig. 4. Average mutual information, $I(X;Y)$, versus *a priori* signal probability, q, with $K_{\mathrm{s}} = 20$, $K_{\mathrm{n}} = 10$.

on the dead time of the receiver. Also, it can be observed from this figure that when $\delta = 0.001$, for $K_{\mathrm{s}} \geq 20$, the proposed capacity metric reaches it maximum value of 1 for both hard and soft decision cases and the limiting effect of dead time vanishes. This happens at a higher value ($K_{\mathrm{s}} \geq 100$) when larger dead time ratio ($\delta = 0.1$) is considered. This trend suggests that by choosing a proper incoming photon rate in practice, the information capacity of the SPAD receiver can be maximized.

The effect of SPAD background counts is shown in Fig. 6 where $K_{\mathrm{s}} = 100$ and $\delta = 0.1, 0.001$. According to this figure, when $\delta = 0.001$, background counts with mean counts lower than 100 have a negligible impact on the capacity. However, as the average background count increases, the capacity is severely limited. For larger values of the dead time ratio, the limiting effect of background counts appears in lower values of mean background count. Again, it is seen that higher values of dead time ratio lead to a larger gap between the soft decision and the hard decision capacities. Furthermore, this figure depicts that if the dead time ratio is small ($\delta = 0.001$), keeping the average background count lower than 110 photons/s, will help overcome the limiting effect of dead time on the capacity. However, for larger dead time ratio values such as $\delta = 0.1$, the maximum capacity is achieved only if $K_{\mathrm{n}} \leq 4$ for hard decision case and $K_{\mathrm{n}} \leq 10$ for soft decision case.

In Fig. 7, the capacity curves are plotted as a function of dead time ratio where $K_{\mathrm{n}} = 1$ and $K_{\mathrm{s}} = 10, 100$ are considered. As seen in this figure, dead time ratio values larger than 0.1 severely degrade the capacity. Hence, it is of greatest importance to keep dead time as small as possible.

Figs. 5–7 are presenting capacity results as a function of three parameters: K_{s}, K_{n}, and δ. In each figure, the effect of one parameter is studied while the two other parameters are fixed. In order to optimize the operating conditions and structure of a SPAD receiver to achieve maximum capacity, the effect of these parameters should be treated simultaneously.

V. Conclusions

In this study, the input-output information transfer rate of AQ SPAD receivers is investigated. The AQ SPAD receiver is modeled as a DMC with binary inputs, and a channel capacity metric is proposed to study the effect of SPAD dead time and background counts on the SPAD information transfer rate for both soft decision and hard decision categories. It is identified that soft decision outputs improve the capacity compared with the hard decision case. The proposed channel capacity metric is suitable for theoretical optimization of the device structure and operating conditions to maximize the achievable data rate. Future research will consider the effects of afterpulsing and timing jitter on the capacity of SPAD receivers, which may provide valuable insights into the fundamental limits of SPAD receivers for optical communication systems.

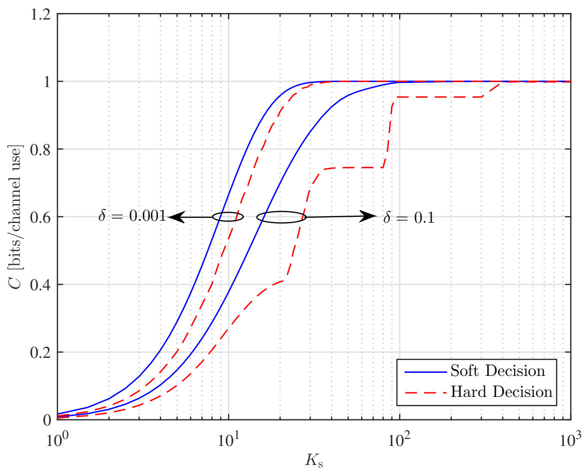


Fig. 5. Capacity of the SPAD receiver as a function of average signal count, K_{s}, with $K_{\mathrm{n}} = 10$ and $\delta = 0.1$, and 0.001.

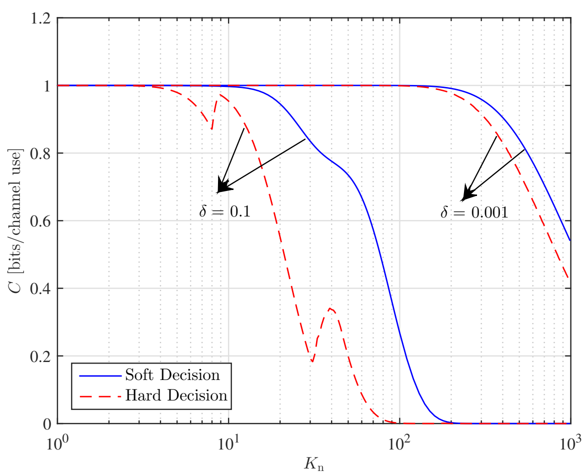


Fig. 6. Capacity of the SPAD receiver as a function of average background count, K_{n}, with $K_{\mathrm{s}} = 100$ and $\delta = 0.1$, and 0.001.

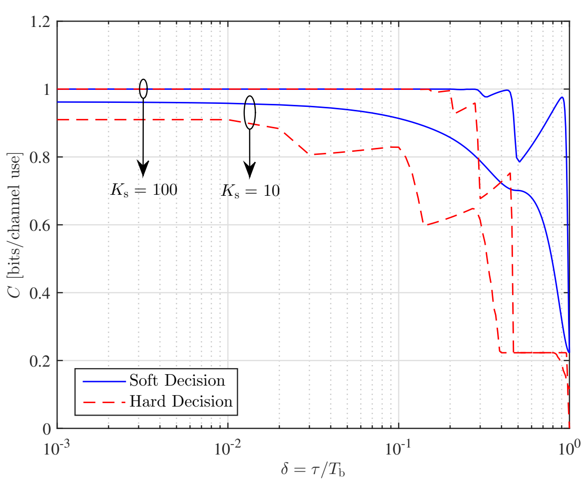


Fig. 7. Capacity of the SPAD receiver as a function of dead time ratio, δ, with $K_{\mathrm{n}} = 1$ and K_{s} =10, and 100.

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